1.Sol

22MA201 & ENGINEERING ~~CHEMISTRY~~ MATHEMATICS - II

$\vec{r}(t) = t\hat{i} + t^2\hat{j}$

velocity vector, $\vec{v}(t) = \frac{d\vec{r}}{dt}$

$\vec{v}(t) = \hat{i} + 2t\hat{j}$

Interpretation:

* constant motion in x direction, Increasing motoin in y-direction
* The path is parabolic

2.

Scalar line Integral	Vector line Integral
* Integrates a scalar function over a curve $\int_c f\,ds$	* Integrates a vector field along a curve $\int_c \vec{F}\cdot d\vec{r}$
* magnitude only	* Magnitude and direction

3.Sol $\vec{F} = (x+y)\hat{i} + (x-y)\hat{j}$ —①

Green's Theorem:

$$\oint_C M(x,y)\,dx + N(x,y)\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA$$

$\therefore M = x+y$, $N = x-y$ [From eqn ①

$\frac{\partial N}{\partial x} = 1 \quad \frac{\partial M}{\partial y} = 1$

$$\oint_C M(x,y)\,dx + N(x,y)\,dy = \iint_R \left(\frac{dN}{dx} - \frac{\partial M}{\partial y}\right) dA$$

$$= \iint_R (1-1)\,dA = \iint_R 0\,dA$$

$$= 0$$

4. Sol

A positive value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} > 0$ indicates positive circulation (counterclockwise rotation) of the field

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5. The line integral of a force feild along a curve gives the work done by the force on the particle

$$\int_C \vec{F} \cdot d\vec{r}$$

Interpretation:
It depends on both the force and the path followed

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6. Path ~~indepeden~~ independence means the work done depends only on the start and end points, not on the path followed.

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7. Sol

$$\vec{F} = y\hat{i} + x\hat{j}$$

$$P = y, \quad Q = x$$

using Green's theorem integrant:

$$\oint_C M(x,y)dx + N(x,y)\,dy = \iint_R \left(\frac{dQ}{dx} - \frac{dP}{dy}\right)dA$$

$$= \iint_R (1-1)\,dA = \iint_R 0\,dA$$

$$= 0$$

Hence, It is indicating there is no circulation or rotation in the fluid flow.

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8. Sol

$$\vec{F} = (x+y)\hat{i} + (2x-y)\hat{j}$$

$P = x+y$
$Q = 2x-y$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial(2x-y)}{\partial x} - \frac{\partial(x+y)}{\partial y}$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2-1 = 1$$

$\therefore$ The required value is 1

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9. <u>sol</u>

To evaluate : $\iint_S \vec{F}\cdot\hat{n}\,ds$

Given : $\vec{F} = x^2\hat{i} + yz\hat{j} + z^2\hat{k}$ $\quad \hat{n} = \hat{j}$ $\quad y=1$, $0 \le x \le 2$, $0 \le z \le 1$

$\vec{F}\cdot\hat{n} = (x^2\hat{i} + yz\hat{j} + z^2\hat{k})\cdot\hat{j} = yz \Rightarrow$ For $y=1$, $\vec{F}\cdot\hat{n} = z$

$ds = dx\,dz$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = \int_0^2\int_0^1 z\,dz\,dx = \int_0^2 \left[\frac{z^2}{2}\right]_0^1 dx = \int_0^2 \frac{1}{2}\,dx$$

$$= \left[\frac{x}{2}\right]_0^2 = \left[\frac{2}{2} - \frac{0}{2}\right]$$

$$= 1$$

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10. <u>sol</u>

Stroke's Theorem:

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S \nabla\times\vec{F}\cdot\hat{n}\,ds \quad \text{—①}$$

Given:

$\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$ $\quad x=0$, $0 \le y \le 1$, $0 \le z \le 1$

For yz plane, unit normal vector, $\hat{n} = \hat{i}$

$ds = dy\,dz$

$$\nabla\times\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z & x & y \end{vmatrix} = \left(\frac{\partial y}{\partial y} - \frac{\partial x}{\partial z}\right)\hat{i} + \left(\frac{\partial z}{\partial z} - \frac{\partial y}{\partial x}\right)\hat{j} + \left(\frac{\partial x}{\partial x} - \frac{\partial z}{\partial y}\right)\hat{k}$$

$$= \hat{i} + \hat{j} + \hat{k}$$

$(\nabla\times\vec{F})\cdot\hat{n} = (\hat{i}+\hat{j}+\hat{k})\cdot(\hat{i}+0\hat{j}+0\hat{k})$

$(\nabla\times\vec{F})\cdot\hat{n} = 1$

From eqn (i)

$$\int_C \vec{F}\cdot d\vec{r} = \iint_S \nabla\times\vec{F}\cdot\hat{n}\,ds$$

$$= \int_0^1\int_0^1 1\,dy\,dz = \int_0^1 [y]_0^1 dz = \int_0^1 1\,dz = [z]_0^1$$

$$\int_C \vec{F}\cdot d\vec{r} = 1$$

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11. Sol

To Evaluate: $\iint_S \vec{F}\cdot\hat{n}\,ds$

$x=2$

$0\le y\le 1$, $0\le z\le 2$

$\therefore \hat{n}=\hat{i}$

$\vec{F} = 2xy\hat{i} + yz\hat{j} + xz\hat{k}$

$\vec{F}\cdot\hat{n} = (2xy\hat{i}+yz\hat{j}+xz\hat{k})\cdot(\hat{i}+0\hat{k}+0\hat{j})$

$\vec{F}\cdot\hat{n} = 2xy$

$x=2$

$\therefore \vec{F}\cdot\hat{n} = 4y$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = \int_0^2\int_0^1 4y\,dy\,dz = \int_0^2 4\left[\frac{y^2}{2}\right]_0^1 dz = \int_0^2 2\,dz = 2[z]_0^2$$

$= 2(2-0)$

$= 4$ //

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12. Sol

Stroke's Theorem:

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S \nabla\times\vec{F}\cdot\hat{n}\,ds$$

Given:

$\vec{F} = xy\hat{i} + yz\hat{j} + 2x\hat{k}$

$y = 0$
$0 \le x \le 1$
$0 \le z \le 1$

$\therefore \hat{n} = \hat{j}$

$ds = dx\,dz$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & yz & 2x \end{vmatrix}$$

$$= \left(\frac{\partial(2x)}{\partial y} - \frac{\partial(yz)}{\partial z}\right)\hat{i} + \left(\frac{\partial(xy)}{\partial z} - \frac{\partial(2x)}{\partial x}\right)\hat{j} + \left(\frac{\partial(yz)}{\partial x} - \frac{\partial(xy)}{\partial y}\right)\hat{k}$$

$$= (0-y)\hat{i} + (0-z)\hat{j} + (0-x)\hat{k}$$

$$\nabla \times \vec{F} = -y\hat{i} - z\hat{j} - x\hat{k}$$

$$(\nabla \times \vec{F})\cdot\hat{n} = (-y\hat{i} - z\hat{j} - x\hat{k})\cdot(0\hat{i} + \hat{j} + 0\hat{k})$$

$$= -z$$

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_S (-z)\,ds = \int_0^1\int_0^1 (-z)\,dx\,dz$$

$$= \int_0^1 -z[x]_0^1\,dz = \int_0^1 -z\,dz = -\left[\frac{z^2}{2}\right]_0^1 = -\frac{1}{2}$$

$$\oint_C \vec{F}\cdot d\vec{r} = \frac{-1}{2}$$

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13. Sol

Green's theorem:

$$\oint_C M(x,y)dx + N(x,y)dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)dA$$

Given

$\vec{F} = y\hat{i} + x\hat{j}$

$0 \le x \le 1$, $0 \le y \le 1$

Here

$N = x$

$M = y$

$$\frac{\partial N}{\partial x} = 1 \qquad \frac{\partial M}{\partial y} = 1$$

$$\oint_C \vec{F}\cdot d\vec{r} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA$$

$$= \iint_R (1-1)\, dA = \iint 0\, dA$$

$$= 0$$

The work done is zero

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14. Sol

Divergence Theorem:

$$\iint_S \vec{F}\cdot\hat{n}\, ds = \iiint (\nabla\cdot\vec{F})\, dv$$

$$\vec{F} = (x^2-yz)\hat{i} + (y^2-xz)\hat{j} + (z^2-xy)\hat{k}$$

$0 \le x \le a$, $\quad 0 \le z \le c \qquad dv = dz\,dy\,dx$

$0 \le y \le b$,

$$\nabla\cdot\vec{F} = \frac{\partial}{\partial x}(x^2-yz) + \frac{\partial}{\partial y}(y^2-xz) + \frac{\partial}{\partial z}(\cancel{x}^2-xy)$$

$$= 2x + 2y + 2z$$

$$\iint_S \vec{F}\cdot\hat{n}\, ds = \iiint_V (2x+2y+2z)\, dv$$

$$= 2\iiint_V (x+y+z)\, dv$$

$$= 2\int_0^a\int_0^b\int_0^c (x+y+z)\, dz\, dy\, dx$$

$$= 2\left[\frac{a^2bc}{2} + \frac{ab^2c}{2} + \frac{abc^2}{2}\right]$$

$= a^2bc + ab^2c + abc^2$

$\iint_S \vec{F}\cdot\hat{n}\,ds = abc(a+b+c)$

Interpretation:

since the flux value is positive

$abc(a+b+c) > 0$

there is a net outward flow of heat from the rectangular region

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15. Sol

Given:

$\vec{F} = 2x\hat{i} + y\hat{j} + z\hat{k}$

$x = 1,\ 0 \le y \le 2,\ 0 \le z \le 1$ $\quad \therefore \hat{n} = \hat{i}$

To find: $\iint_S \vec{F}\cdot\hat{n}\,ds$.

$\vec{F}\cdot\hat{n} = (2x\hat{i} + y\hat{j} + z\hat{k})\cdot(\hat{i})$

$= 2x$

$\iint_S \vec{F}\cdot\hat{n}\,ds = \int_0^1\int_0^2 2\,dy\,dz = \int_0^1 2[y]_0^2\,dz = 2\int_0^1 [2-0]\,dz$

$= 4\int_0^1 dz = 4[z]_0^1 = 4(1-0)$

$= 4$ //

$\therefore$ The total flux through the surface is 4 units

16. Sol

$\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ $\quad dV = dx\,dy\,dz$

$0 \le x, y, z \le 1$

Divergence Theorem: $\iint_S \vec{F}\cdot\hat{n}\,ds = \iiint_V (\nabla\cdot\vec{F})\,dV$

$$\nabla\cdot\vec{F} = \frac{\partial(4xz)}{\partial x} + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(yz)$$

$$= 4z - 2y + y$$

$$\nabla\cdot\vec{F} = 4z - y$$

$$\therefore \iint_S \vec{F}\cdot\hat{n}\,ds = \iiint_V (4z-y)\,dV = \iiint_V (4z-y)\,dx\,dy\,dz$$

$$= \int_0^1\int_0^1\int_0^1 (4z-y)\,dx\,dy\,dz$$

$$= \int_0^1\int_0^1 (4z-y)\,dy\,dz = \int_0^1 (4z-y)\,dz$$

$$= \left[2z^2 - \frac{z}{2}\right]_0^1 = 2 - \frac{1}{2}$$

$$\iint_S \vec{F}\cdot\hat{n}\,ds = \frac{3}{2}$$

Interpretation:

Since the flux is positive

$\frac{3}{2} > 0$

There is a net outward heat flow from the cubic region

17. Sol

$f(z) = z^2$

$f(z) = (x+iy)^2$

$f(z) = x^2 - y^2 + 2ixy$

CR equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x} = 2, \quad \frac{\partial v}{\partial y} = 2x, \quad \frac{\partial u}{\partial y} = -2y, \quad \frac{\partial v}{\partial x} = 2y$$

$\therefore$ It obeys CR equations

Therefore, $f(z) = z^2$ is differentiable at $z = 1+i$

18. Sol $w = z^2$

* For each complex number z, there is only one unique value of w. Hence, the function does not produce multiple outputs for the same input

* $\therefore$ $w = z^2$ is a single valued function

19. Sol Given: $v(x,y) = 2xy$

To find: $u(x,y)$

Milne-Thompson method:

$$f'(z) = \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}$$

$$\frac{\partial v}{\partial y} = 2x, \quad \frac{\partial v}{\partial x} = 2y$$

$$f'(z) = 2x + i2y = 2(x+iy)$$

$$f'(z) = 2z \qquad [\because z = x+iy$$

$\therefore f(z) = \int 2z \, dz$

$f(z) = z^2 + C$

$z^2 = (x+iy)^2$

$z^2 = \underbrace{x^2 - y^2}_{u} + i\underbrace{(2xy)}_{v}$

$\therefore \boxed{u(x,y) = x^2 - y^2}$

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20. Sol

Given : $u(x,y) = x^2 + 2xy$

To find : $f'(z)$

Using Milne - Thomson Method,

~~$f'(z) = u_x(z,0) - i(z$~~

$f'(z) = u_x(z,0) - iu_y(z,0)$

$u_x = 2x + 2y$

$u_y = 2x$

put $y=0$, $x=z$

$\therefore u_x = 2z$

$u_y = 2z$

$\therefore f'(z) = 2z - i(2z)$ $\boxed{f'(z) = 2z(1-i)}$

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21. Sol

* If complex function $f(z)$ is differentiable, then it is continuous
* But a continuous function need not be differentiable
* Hence, Differentiability $\Rightarrow$ continuity
* But the converse is not always true

22. Sol

$w = \sqrt{z}$

* For one value of z, the function gives two values of w (positive and negative roots)

* So, $w = \sqrt{z}$ is multi-value function

23. Sol

Given: $u(x,y) = e^{x}\cos y$

To find: $f(z)$

Milne-Thomson method:

$$f'(z) = u_x - iu_y$$

$$u_x = e^{x}\cos y$$

$$u_y = -e^{x}\sin y$$

$$f'(z) = e^{x}\cos y + ie^{x}\sin y$$

$$= e^{x}(\cos y + i\sin y)$$

$$f'(z) = e^{x+iy} = e^{z} \Rightarrow f'(z) = e^{z}$$

$$f(z) = \int e^{z}dz$$

$$\boxed{f(z) = e^{z} + C}$$

24. Sol

Given: $v(x,y) = x^2 - y^2$

To find: $f'(z)$

Milne-Thomson method:

$$f'(z) = v_y(z,0) + iv_x(z,0)$$

$$v_x = 2x$$

$$v_y = -2y \Rightarrow f'(z) = 0 + i(2z)$$

Put $y=0$, $x=z$

$\therefore v_x = 2z$ $\quad \boxed{f'(z) = 2iz}$

$v_y = 0$

25. Sol

$f(z) = e^z$ WKT $z = x + iy$

$\therefore e^z = e^x(\cos y + i\sin y)$

cauchy - Riemann eqn (CR eqn):

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial u}{\partial x} = e^x \cos y$$

$$\frac{\partial v}{\partial y} = e^x \cos y$$

$$\frac{\partial u}{\partial y} = -e^x \sin y$$

$$-\frac{\partial v}{\partial x} = -e^x \sin y$$

$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

* It satisfy CR eqn. Hence, e^z is analytic fuction in the entire function,

* significance in communication:

↳ complex exponential functions are used to represent amplified and oscillating signals in communication systems, modulation and signal processing

26. Sol

$w = (1+i)z$

let

$z = x + iy$ [assuming

$w = u + iv$

$w = (1+i)z$

$= (1+i)(x+iy)$

$w = (x-y) + i(x+y)$

$\quad\ \ u \qquad\qquad v$

$u = x-y$ ①, $v = x+y$ ②

given : $y > 2$

eqn ② - eqn ① $\Rightarrow$ $v-u = (x+y)-(x-y)$

$v-u = 2y$

$\frac{v-u}{2} = y$

$\therefore \frac{v-u}{2} > 2$

$v-u > 4 \Rightarrow \boxed{v > u+4}$

The image in the w-plane is $v > u+4$

Transform of characteristics:

- Rotation by 45°
- straight lines remain straight lines

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28. Sol $f(z) = (x^2-y^2+2x) + i(2xy+2y)$

Here

$u = x^2-y^2+2x \qquad v = 2xy+2y$

$\frac{\partial u}{\partial x} = 2x+2 \qquad \frac{\partial v}{\partial y} = 2x+2$

$\frac{\partial u}{\partial y} = -2y \qquad \frac{\partial v}{\partial x} = 2y$

It satisfy CR eqn; $\frac{\partial u}{\partial x} = 2x+2 = \frac{\partial v}{\partial y}\ 2x+2$ and

$\frac{\partial u}{\partial y} = \frac{-\partial v}{\partial x} = -2y$

Hence, $f(z)$ is analytic everywhere

Physical meaning:

The temperature potential is smooth and continuous, indicating a stable heat flow without discontinuities in the thermal field.

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28. Sol

Given: $w = z + (1+4i)$, $|z| = 2$ — ①

we know that, $|z-a| = r$ where $a \Rightarrow$ center, $r \Rightarrow$ radius

$\therefore$ * centre $= (0,0)$ $\quad$ ∵ $c = 0 + 0i$

* radius $= 2$

2) After transformation

$$w = z + (1+4i)$$

∴ the circle shifted by $(1,4)$

* New center $= (1,4)$

* Radius $= 2$ (same)

Result:

Hence the image is $|w - (1+4i)| = 2$

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29. Sol

Given: $v(x,y) = 2xy$

Milne-Thomson method:

$$f'(z) = \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}$$

$\frac{\partial v}{\partial y} = 2x$, $\frac{\partial v}{\partial x} = 2y$

Put $x = z$, $y = 0$ [According to Milne-Thomson method

$\therefore \frac{\partial v}{\partial y} = 2z$, $\frac{\partial v}{\partial x} = 0$

$$f'(z) = \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}$$

$$f'(z) = 2z$$

we want $f(z)$, so integrate it

$$f(z) = \int 2z\,dz$$

$$\boxed{f(z) = z^2 + c}$$

$\therefore$ The required analytic function is $f(z) = z^2 + c$ //

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30. Sol

Given points: $z_1 = 1$, $z_2 = i$, $z_3 = -1$

mapped onto : $w_1 = 0$, $w_2 = 1$, $w_3 = -1$

using the bilinear transformation formula:

$$\frac{(w-w_1)(w_2-w_3)}{(w-w_3)(w_2-w_1)} = \frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}$$

subt values:

$$\frac{(w-0)(1+1)}{(w+1)(1-0)} = \frac{(z-1)(i+1)}{(z+1)(i-1)}$$

$$\frac{2w}{w+1} = \frac{(z-1)(1+i)}{(z+1)(i-1)}$$

$$\frac{2w}{w+1} = -i\frac{(z-1)}{(z+1)} \quad \left[\text{where } \left(\frac{1+i}{i-1}\right) = -i\right]$$

$$2w(z+1) = -i(z-1)(w+1)$$

$$2wz + 2w = -i(zw + z - w - 1)$$

$$2wz + 2w + iwz - wi = -iz + i$$

$w(2z + 2 + iz - i) = -iz + i$

$$w = \frac{-i(z-1)}{2(z+1) + i(z-1)}$$

Result:

The required bilinear formation is $w = \dfrac{-i(z-1)}{2(z+1)+i(z-1)}$

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31. Sol

$u(x, y) = x^2 - y^2$

1) If $u_{xx} + u_{yy} = 0$ then it is harmonic function

$$u_{xx} = \frac{d^2}{dx^2}(x^2 - y^2) = \frac{d}{dx}\left(\frac{d}{dx}(x^2 - y^2)\right)$$

$$= \frac{d}{dx}(2x)$$

$$u_{xx} = 2$$

$$u_{yy} = \frac{d^2}{dy^2}(x^2 - y^2) = \frac{d}{dy}\left(\frac{d}{dy}(x^2 - y^2)\right)$$

$$= \frac{d}{dy}(-2y)$$

$$u_{yy} = -2$$

$$u_{xx} + u_{yy} = 0$$

$\therefore$ It is harmonic

2) CR eqn:

$$\frac{du}{dx} = \frac{dv}{dy}, \quad \frac{du}{dy} = -\frac{dv}{dx}$$

$$\frac{du}{dx} = \frac{dv}{dy} = 2x, \quad \frac{du}{dy} = -\frac{dv}{dx} = -2y$$

Harmonic conjugates is $v(x, y)$

$$v(x, y) = \int \frac{dv}{dx}$$

$$\boxed{v(x,y) = 2x + c}$$

It is the required harmonic conjugate where $u(x,y) = x^2 - y^2$

32. Sol

Given points: $z_1 = 0, z_2 = i, z_3 = -i$

mapped onto: $w_1 = 1, w_2 = -1, w_3 = 0$

Using bilinear transformation formula:

$$\frac{(w-w_1)(w_2-w_3)}{(w-w_3)(w_2-w_1)} = \frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}$$

$$\frac{(w-1)(-1-0)}{(w-0)(-1-1)} = \frac{(z-0)(i+i)}{(z+i)(i-0)}$$

$$\frac{-(w-1)}{-2w} = \frac{2zi}{i(z+i)} \Rightarrow \frac{w-1}{2w} = \frac{2z}{z+i}$$

$$(w-1)(z+i) = (2w)(2z)$$

$$wz - z + iw - i = 4wz \Rightarrow wz + iw - 4wz = z +$$

$$\cancel{w(z+i-4z} \quad w = \frac{iz-1}{iz-4z-1}$$

Engineering significance:

*Bilinear transformation is used in signal processing and control systems to transform systems while preserving stability